



CONTACT

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EDUCATION

2024 → PRESENT
TAMPERE UNIVERSITY

- PhD candidate
- Topic: Signal Processing and Applications of SPADs for Passive Imaging

2020 - 2022
TAMPERE UNIVERSITY

- Master of Science (Computer Science)
- Major in Signal Processing and Machine Learning

2017 - 2020
TAMPERE UNIVERSITY OF TECHNOLOGY

- Bachelor of Science (Computer Science)
- Major in Software Development and Machine Learning

SKILLS

- 6+ years of working experience in computer vision and ML
- C++, Python
- Matlab
- PyTorch and Tensorflow

LANGUAGES

- Finnish (Native)
- English (Fluent)
- Swedish (Satisfactory)

ALEKSI SUONSIVU

SENIOR ALGORITHM ENGINEER

PROFILE

Senior Algorithm Engineer & PhD Candidate with 6+ years of industry experience specializing in deep learning for image restoration and multi-modal sensor fusion. Proven track record of balancing full-time algorithm development with advanced academic research. An extroverted problem-solver who thrives in collaborative environments.

WORK EXPERIENCE

- **Agate Sensors Oy** 2026 - PRESENT
Senior Algorithm Engineer
 - Development of spectral imaging algorithms
- **Huawei Technologies Finland** 2020 - 2026
Camera Algorithm Engineer
 - Created deep learning solutions for image restoration using CMOS, Event, and SPAD sensors.
 - Advanced passive SPAD imaging through novel pipelines, simulation tools, and published research.
 - Led real-world validation through calibration, dataset capture, and image quality evaluation.
- **Teleste** 2019 - 2020
Summer Trainee / Freelance
 - Embedded system and software testing
 - Hardware repair

SELECTED PUBLICATIONS

- **Color Preserving CMOS-SPAD Fusion for Multi-Frame HDR**
A. Suonsivu, L. Salmela, L. Helin, L. Uosukainen, G. Boracchi
Proceedings of the Winter Conference on Applications of Computer Vision 2026
- **Accurate Simulation Pipeline for Passive Single-Photon Imaging**
A. Suonsivu, L. Salmela, L. Uosukainen, E. Peretti, RC. Bilcu, G. Boracchi
IEEE Sensors Journal
- **Dark Count Removal in Photon-Counting SPAD Arrays**
E. Peretti, A. Suonsivu, L. Salmela, L. Uosukainen, RC. Bilcu, G. Boracchi
2025 IEEE International Conference on Image Processing (ICIP), 522-527

INTERESTS AND VOLUNTARISM

- Lead singer / guitarist @ RÄTBOX
- Skateboarding, golf and playing music
- Culture journalist - Wappulehti Tamppi (2019 - 2022)
- Fuksicapitain - Guild of Automation ry (2018 - 2019)

REFERENCE

Eero Salmelin

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